Research Proposal

Funding Credit Lines: Revisiting the Liquidity Synergy Hypothesis

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1 Objective

- **Research question:** Does deposit funding lead to more credit line supply relative to term loans?

In this paper, we use loan-level data to study how exogenous deposit/capital structure shocks affect the supply of credit commitments. This setting allows us to directly observe adjustments in credit limits, utilization, and pricing, providing a more stringent test of the deposit–credit line synergy mechanism.

1.1 Rationale

Term loans and credit lines are different types of credit products. Importantly, they have two key differences in terms of how they are funded, which have implications for their supply. First, term loans require cash-on-hand (or, at least, balance sheet space for the simultaneous creation of deposits), while credit lines do not. Second, credit lines are demandable, so their provision exposes the supplier to liquidity risk.[1]

According to the liquidity synergy hypothesis (Kashyap et al., 2002), the second characteristic of credit lines implies that deposits have a special role in their provision, since deposits are also demandable.

Therefore, the supply of credit via term loans or credit lines might behave differently in response to a change in deposits.

2 Motivation

- Are deposits different from other sources of funding? How important are they for the provision of credit lines?
  - Rise of stablecoins: if banks lose deposits to stablecoins, what happens to credit lines?
  - Decline in deposits as a share of household wealth (Buchak et al 2025) and the rise of MMFs: if banks lose deposits to MMFs, what happens to credit lines?
  - Rise of shadow banks: Why hasn't the share of shadow bank lending increased as much in the credit line segment? Why do they rely on bank credit lines?

[1] We define credit lines as demandable credit facilities. The borrower has access to credit up to a limit and can draw down on this credit whenever he wants. In that sense, the loan amount is decided by the borrower. The distinctive feature of a credit line is the existence of an "unused" portion of the credit facility.

- Transmission of financial shocks
  - Monetary policy → deposit flows → lending (credit lines)
  - From credit supply to real effects:
    - Some firms are more dependent on credit lines vs term loans (Campello, Greenwald, etc)
- Why do we care about credit lines?
  - Relevant credit product (as a share of total credit) - Greenwald et al. (2025) says credit lines are more than 50% of used credit
  - Unused credit is large - Greenwald et al. (2025) says it is larger than the used portion of credit lines
- Testing a theory: Liquidity Synergy hypothesis
  - Are there economic synergies between deposit-taking and credit commitment provision?
  - Information synergies vs Liquidity synergies
- Extra
  - Firm behavior
    - What kind of firm has a harder time substituting between banks and credit types?
    - How much insurance do credit lines actually provide?
    - Does substitution matter much for credit supply identification? Khwaja and Mian (2008) assumes it away - methodological discussion (Bergman et al, 2026 - NBER)

3 Literature

The Liquidity Synergy hypothesis proposed by Kashyap et al. (2002) says that as long as deposit withdrawals and credit line drawdowns are not perfectly correlated, banks can save on costly liquidity buffers by pooling those two activities together.

Yet, the existing literature directly addressing the interaction between deposits and credit commitments remains relatively limited. Gatev et al. (2009) and Gatev and Strahan (2006) are the main contributors in favor.

More recent work, enabled by improved data availability, has largely shifted toward a household finance perspective, focusing on the role of credit commitments as insurance and on how households respond to changes in credit line availability. A key insight from this literature is the substantial variation in credit commitments, driven by both supply- and demand-side factors.

Some papers bring evidence that contradicts the liquidity synergy hypothesis. Chava et al (2023) and Darst et al (2025)

4 Approach

1. Reduced form regressions with aggregated data
    - Replicate KRS with improved measurements
    - Extend their results: More years, bank fixed-effects, include credit cards
2. Credit supply identification with loan-level data
    - Khajawa and Mian (2008)
    - Exogenous deposit shock
3. Model? – I think a model would be helpful to clarify what the control variable of banks is, and what ratios I should be looking at (transaction deposit to total deposits; deposit to assets; etc). And to make explicit what is distinct about credit line (vs term loans)

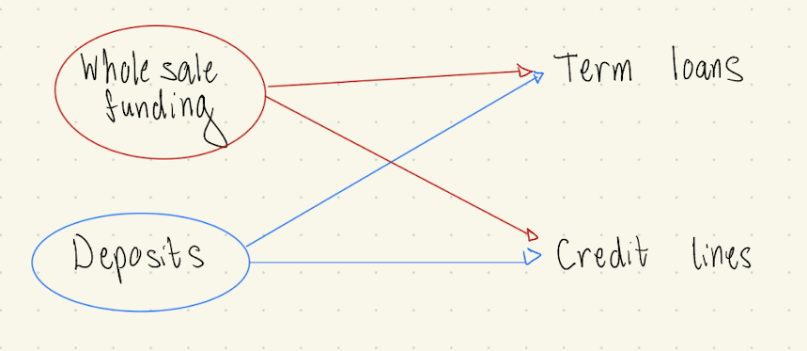
4.1 Bank-level regression

$$\frac{C_b}{C_b + T_b} = \beta_0 + \beta \frac{D_b}{A_b} + \epsilon_b$$

where C_b is credit commitment by bank b, T_b is term loans, D_b is transaction deposits, A_b is assets.

The coefficient of the regression above should not be interpreted as a causal effect for multiple reasons:

- Banks have control over both dependent and independent variables, so bank unobserved characteristics will confound the estimation
    - Business model sorting - e.g. regional banks/credit unions have a lot of deposits and do a lot of term lending (mortgages and auto loans)
    - Client composition - e.g. banks that have more corporate clients might have a higher proportion of credit lines committed and a lower amount of transaction deposits
    - Risk appetite
    - Management quality
    - Deposit market power
    - Marketing and intangibles (customer capital)



5 Data

5.1 Peruvian Credit Registry

- Source: Superintendencia de Banca, Seguros y AFPs (SBS)
- Coverage: Universe of loans made by financial institutions
  - Firm side: ≈ 3.4M borrowers
  - Consumer side: ≈ 3.7M borrowers
- Period: September 2010 - December 2023
- Frequency: monthly
- Granularity: loan level
- Variables: date; firm id; bank code; balance outstanding; unused credit; type of credit; currency; delinquency (categorization of the overdue balance)

Information about the firm: id; number of banking relations; total credit; size (categorical); firm type (foreign, public or other); what else

5.2 Classification of Credit Lines

Product type + Revolving

5.3 Comparison to Other Datasets

Main advantages of the credit register:

- universe of loans
- observe both credit limits and usage

- individual id - allows link between different accounts of the same individual
- both corporate and consumer loans

Other datasets I've seen in the literature:

- Consumer Credit Panel from Fed NY – only has aggregate credit limit (all accounts of the same individual)
- Fed Y-14 data on credit cards – account-level data but no individual ID
  - has a variable identifying whether the change in the limit was initiated by the lender or the borrower
- LPC Dealscan – does not cover consumer loans; does not cover all credit lines (?)
- Shared National Credit program (SNC) from the Fed/FDIC
- Equifax data – maybe the most suitable data for the identification, but it covers only credit card lending

6 Empirical Strategy

Our goal is to estimate the effect of deposits on credit supply for term loans and credit lines. We employ the approach from Ivashina et al. (2022), which is an extension of Khwaja and Mian (2008) that allows for the effect on distinct credit types to be different.

$$\Delta L_{fblt} = \alpha_{flt} + \beta_l \Delta D_{bt} + \epsilon_{fblt}$$

where L_{ilbt} is the loan amount that firm i has with bank b in the loan type l, which can be term loans or credit lines, at time t; α_{ilt} is a firm-type of loan fixed effect, which controls for demand shocks; D_{bt} is bank deposits. It still requires an instrument for deposits that should be uncorrelated with bank unobserved characteristics.

6.1 Rationale

This strategy deals with two identification challenges:

1. Simultaneity problem: credit demand
2. Endogeneity problem: bank's choice of deposits

First, the bank has some control over both deposits and loans, so the two can be confounded by changes in unobservable bank characteristics. To overcome this concern, we need a source of variation in deposits that is uncorrelated with bank characteristics. Second, even with this variation, a concern remains that the deposit shock might be correlated with credit demand shocks. This is addressed with a firm fixed effect, which controls for changes in demand under the assumption that demand shocks lead to a scale effect in the credit demand of a firm, that is, the firm does not shift its lending towards any one of the banks with which it has relationships.

6.2 Natural Disaster Instrument

One potential source of variation for the deposits we considered using is the instrumental variable from Kundu et al. (2021) which combines the variation in banks' exposure to the local deposit market with natural disasters (shift-share instrument).

$$\Gamma_{bt} = \sum_c \frac{D_{b,c,t-1}}{\sum_c D_{b,c,t-1}} \varepsilon_{ct}$$

Γ_{bt} is the bank deposit shock for bank b at time t constructed by averaging the disaster shocks ε_{ct} of each county c by the bank-county deposit share.

For the instrument to be valid, we need the exclusion and relevance assumptions to hold.

For the exclusion assumption, the shares need to be uncorrelated with the unobserved confounder of loans and deposits, after controlling for the firm fixed effect. Otherwise, the exogeneity of the natural disasters is also sufficient for consistency (with asymptotics relying on the number of regions). Ideally, the two conditions hold.

In order for the relevance assumption to hold, we need the local deposit shares to be correlated with the variation in deposits. In economic terms, we can state four conditions that would make this more likely.

1. Sizable natural disaster events
2. Natural disasters negatively affect deposits
3. Banks are geographically concentrated
4. Banks have different exposure to the natural disaster

The first two conditions imply that natural disasters are correlated with local deposits. Figure 1 shows the incidence of natural disasters in Peru. Kundu et al. (2021) provide evidence that natural disasters negatively affect local deposits in the US.

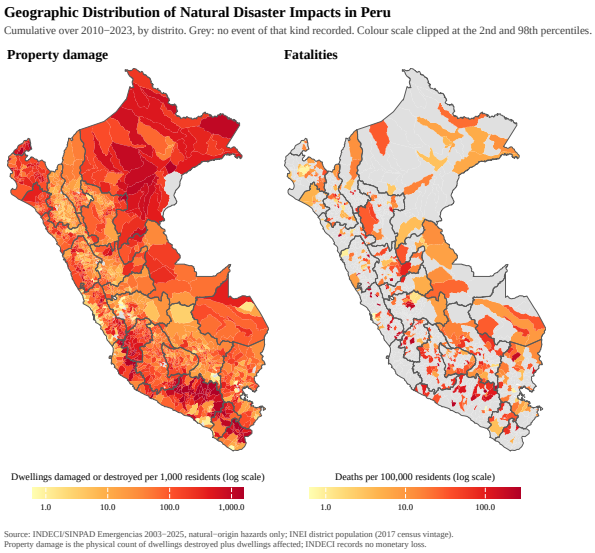


Figure 1

To understand conditions 3 and 4, consider the case where all banks have their deposits evenly distributed across the regions. That would imply constant bank local deposit share, both within bank and across banks. With constant share, the correlation with deposit variation would be zero. Therefore, we would like large variation in shares for relevance to be plausible.

The variation in the shares is higher if the shares of each firm are unevenly distributed and the shares across firms are unaligned. Figure 2 shows that banks are geographically concentrated. 5 out of the 17 banks operate with a single branch, thus in a single district. For the multi-branch banks, the top 3 districts account for 58.9% of deposits on average (the median bank has deposits in 50 districts).

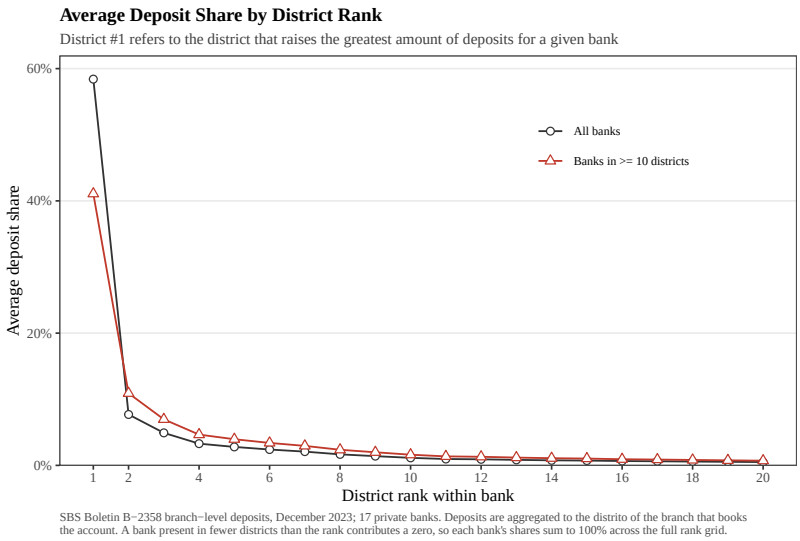


Figure 2

Lastly, condition 4 is the main issue in our setting, as banks in Peru are all concentrated in the same region, Lima. Hence, across banks, the shares are aligned - high in Lima and a couple of other states and low elsewhere. Lima concentrates 81% of all deposits and 50% of bank branches (see Figure 4(a)). This is common across all banks - all banks in Peru have their largest deposit district (and 2nd largest) located in Lima. As a result, all banks are mainly exposed to natural disasters that happen in Lima. The homogeneity of the exposure is undesired, as it reduces the identifying variation. Note that this concentration roughly reflects the geographic distribution of the population (Figure 4(b)), so it is unlikely to be a data issue, such as the assignment of deposits to headquarters.

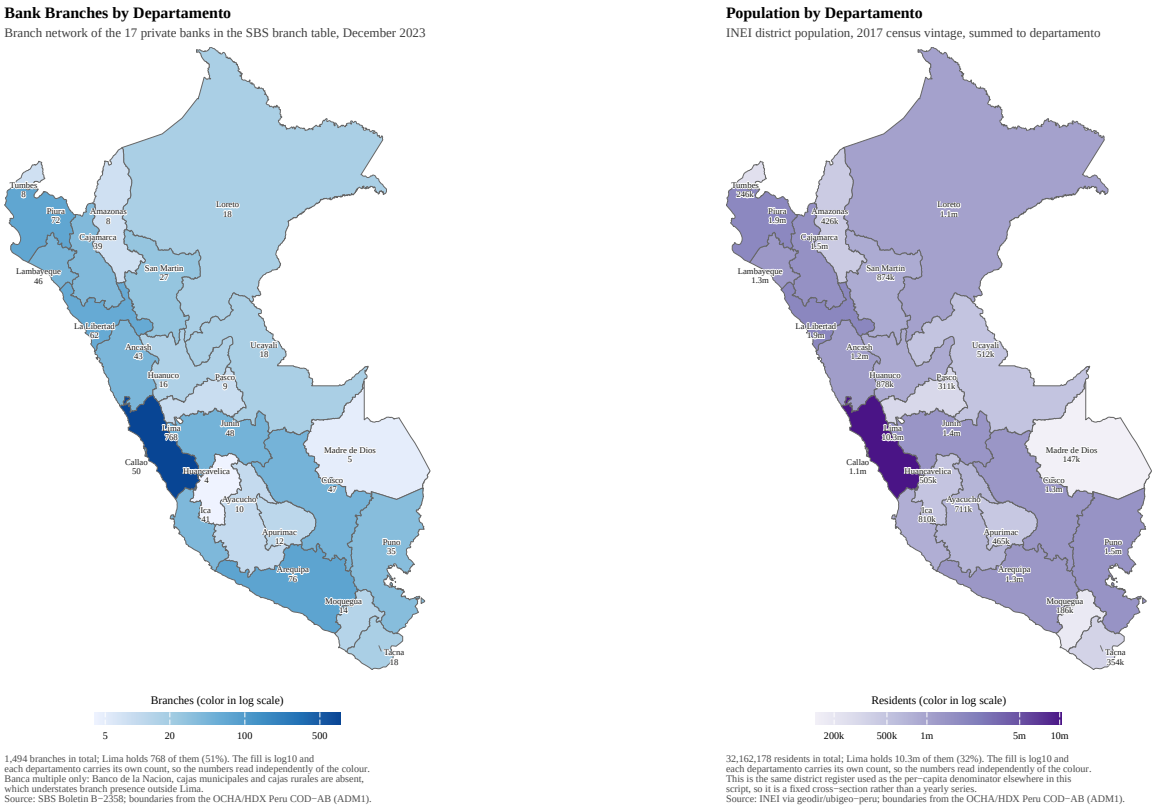
6.3 Reserve Requirement Shock

Changes in reserve requirements also provide a source of variation in banks' access to deposit funding. When required reserves are remunerated below the return on alternative assets, reserve requirements act as an implicit tax on deposits: they raise the cost of financing loans with deposits by reducing the fraction of deposit funding available for lending. There are a few gaps in this argument when we think about credit lines, because the reserves can be drawn down by the bank when there is a deposit withdrawal shock, but not when there is a credit line drawdown shock. I would have to put that in the KRS model and see what it says. The relevant equivalence is thus between reserve requirement increases and reductions in usable deposit funding, rather than necessarily between reserve requirement increases and reductions in the measured stock of deposits.

$$L + R = D + E \Rightarrow L + \varphi D = D + E \Rightarrow L = (1 - \varphi)D + E$$

Peru's reserve requirement framework allows us to exploit this mechanism through a shift-share design. The Central Reserve Bank of Peru sets differentiated requirements across liability categories, including deposits denom-

Figure 3



(a) Branches

(b) Population

inated in domestic and foreign currency, and has adjusted these requirements differently over time. For example, domestic- and foreign-currency deposits have been subject to distinct marginal reserve requirement rates. These policy differences imply that a common regulatory change can generate heterogeneous changes in banks' effective reserve burdens, depending on their initial funding composition. We capture this exposure using the instrument

$$Z_{b,t} = -\sum_{k} s_{b,k,0} \Delta RR_{k,t}$$

where $s_{b,k,t}$ denotes the share of deposit category k in bank b at time t - we use a fixed baseline period -, and $\Delta RR_{k,t}$ denotes the change in the reserve requirement applicable to that category. The negative sign makes the increases in reserve requirements interpretable as negative deposit shocks. Fixing the shares before the policy changes prevents contemporaneous adjustments in deposit composition from mechanically determining the instrument. The identifying variation comes from the interaction between category-specific policy changes and differences in banks' initial exposure, rather than from contemporaneous changes in their funding choices.

The validity of this design can be motivated through two alternative identification arguments. The first relies on the exogeneity of the policy shifts. complete this.

The second identification argument relies on the exogeneity of the baseline deposit shares. complete this.

6.3.1 Reserve Requirement Details

Peru's reserve requirement framework is built around two main rates that the BCRP sets separately for soles and U.S. dollar obligations. The first is the minimum legal rate, a floor on the reserve a bank must hold as a share of its total reservable obligations. In soles it ranged between 4% and 9% over the sample period, while in dollars it has been fixed at 9% since October 2010. Under a single-rate regime, this rate applies to every reservable obligation. Under a base-plus-marginal regime, it operates as a floor.

The second is the marginal rate, which applies only to obligations above a bank-specific base. The base is the bank's average daily reservable obligations in the general regime over a single reference month named in each circular (for instance, circular XXX from September 2021 set the reference month as July 2021 for the domestic currency reserves, and the marginal rate was in force from September 2021 to 2024). Marginal rates are typically far higher than the legal minimum: 25% in soles between 2021 and 2024, and between 20% and 70% in dollars, where it has been 35% since 2018. This design lets the central bank discourage balance sheet growth in a given currency without raising the reserve on existing obligations.

The rules around the base have changed repeatedly across circulars. The reference month has been reset from time to time. Banks without a history, such as new entities, have no base: the circular instead sets a flat rate up to a specified amount and the marginal rate on the remainder, while in mergers the banks' bases are added together. As a result, the average reserve rate a bank faces lies between the floor and the marginal rate, moving toward the latter as obligations grow beyond the base. Because the base rate depends on each bank's own reserve history in the reference month, the average rate cannot be reconstructed from the circulars alone.

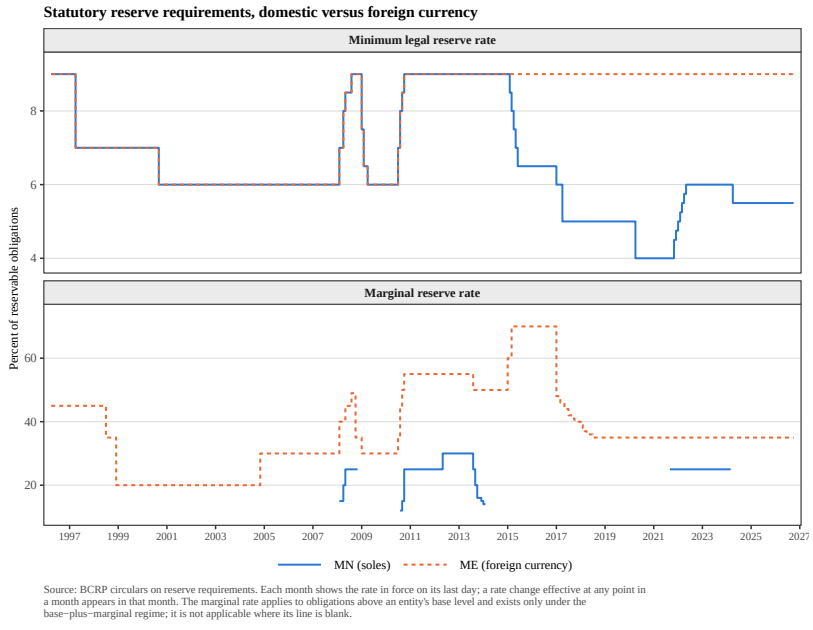


Figure 4

6.4 Remmitances

7 Model

7.1 Simplified KRS

Term loan:

Choose the amount of loans L to maximize profit, under a quadratic cost of external financing e. The bank is a price taker.

$$\begin{aligned} &\max_{L} \; rL - \frac{\gamma}{2}e^2 \\ &\text{s.t.} \quad L = D + e \end{aligned} \tag{1}$$

The solution involves a fixed amount of external financing given the parameters r and γ. Hence, a marginal increase in D is completely used to make additional loans ($dL^*/dD = 1$).

$$\text{FOC: } r - \gamma e = 0 \Rightarrow e = r/\gamma \Rightarrow L = r/\gamma + D$$

Credit line:

Choose the amount of credit commitments C and the liquidity buffer ℓ to maximize profit. Assume constant marginal revenue from the fees charged on the credit line. The cost to the bank is the sum of the opportunity cost of the buffer τ and the cost of raising external financing, with the difference that the additional funds are contingent – only needed if the shocks to the bank are bigger than the liquidity buffer. The bank gets hit with two shocks: ω is a deposit withdrawal shock and z is a credit line drawdown shock.

$$\begin{aligned} &\max_{C,\ell} \; fC - \tau\ell - \frac{\gamma}{2}E[e^2] \\ &\text{where} \quad e = \max\{\omega D + zC - \ell,\ 0\} \end{aligned} \tag{2}$$

The liquidity management problem will interact with the credit supply problem in a way that, under certain conditions, the "marginal propensity of lending" is less than one. To see this, we rewrite the problem as a two step maximization.

$$\begin{aligned} &\max_{\ell} \; \max_{C} \; fC - \tau\ell - \frac{\gamma}{2}E[e^2] \\ &\text{where} \quad e = \max\{\omega D + zC - \ell,\ 0\} \end{aligned} \tag{3}$$

To simplify the problem, we assume the shocks are discrete, $\omega, z \in \{0, 1\}$ with equal probability. We can parametrize the correlation of the shocks by defining $\rho \equiv P(\omega = 1 | z = 1)$.[2]

$$E[e^2] = \frac{\rho}{2}(C + D - \ell)^2 + \frac{1-\rho}{2}(C - \ell)^2 + \frac{1-\rho}{2}(D - \ell)^2$$

[2]Then, $\rho = 1$ implies perfect positive correlation, $\rho = 1/2$ independency, and $\rho = 0$ perfect negative correlation

FOC:

$$C^*(\ell) = 2f/\alpha + \ell - \rho D$$

Substitute this back into the objective to get the conditional value function. Then

$$\ell^* = \frac{(f-\tau)2}{(1-\rho)\alpha} + D$$

Therefore, an increase in D has two effects on credit supply. The direct effect, given a fixed liquidity buffer, is to reduce supply because more deposits "take up" more of the buffer. The indirect effect happens through the change in the liquidity buffer – more deposits make the bank increase the buffer and, as a result, it is optimal to increase commitments. When $\rho < 1$, the indirect channel wins.

8 Possible Issues

Identification challenges:

- Credit demand: are movements in the commitments coming from supply or demand?
- Mechanical increase of deposits: when a bank makes a loan, it creates deposits

identification: deposit shock;

not sure whether we can observe the fees

9 Trying to reason through the idea

Consider the problem of supply of credit lines without the possibility of withdrawing deposits. In this world, there is a set amount of liabilities and the bank chooses how to allocate it between term loans and liquid assets, which allow the bank to supply credit lines. The trade-off in the choice of the size of the liquidity buffer is: more liquid assets reduce the expected cost of external funds in the event of a drawdown, but more liquid assets means giving up the revenue from term loans.

Conjecture: when thinking about credit lines, the Modigliani-Miller benchmark is trivially not realistic. Under MM, there would be no frictions in raising external funds when a drawdown happens, so nothing would limit the banks from offering an infinite amount of credit commitments (if the demand for those commitments was perfectly elastic).

In that world, the source of funding does not affect the bank's choice of loan types. The liquidity synergy hypothesis breaks this by introducing a diversification benefit between deposits and credit lines (capital structure affects the cash flows). Now, the composition of the liability side of the bank BS should affect the asset side.

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Appendix

Details About Credit Commitments

- ☐ fill this part with a general description of how they work (contract details etc)

- 'One feature that is prevalent among credit lines is that they are discretionary in nature. Nearly all credit lines have an "escape clause". This allows banks to withdraw the commitment when it observes "material adverse changes" in the borrower's condition or if a borrower violates a covenant' — Rauf (2023)

The Limitations of KRS

- 5 years of data
- Aggregate data only
- Excludes credit cards
- No fixed-effects (focus on cross-section)